The Rural Care Gap: A Supply-Side Problem

How Specialist Shortages, Not Socioeconomic Barriers, Drive Longer Follow-Up Times for Rural Patients

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Overview

This analysis examines patient journeys through the Stormont Vail Health (SVH) system, focusing on how **geographic rurality** shapes care experiences. By integrating **external data**, patient ZIP codes were first mapped to county-level FIPS codes using a publicly available ZIP-to-FIPS crosswalk dataset (Pilafas, 2024), then matched to the USDA 2023 Rural-Urban Continuum Codes (Economic Research Service, 2024) to assign rurality ratings. These ratings classify every patient and provider location into Metro, Nonmetro, or Rural categories. SVH's service area spans 19 counties in Northeast Kansas, predominantly rural, with care centralized around Topeka. The dataset includes ~200,000 encounters involving 1,384 providers across 382 unique office addresses in 79 cities. Key Finding: The Follow-Up Gap.

A patient journey is defined as the sequence of encounters linked by a common diagnosis; follow-up time is the gap between consecutive encounters within that journey. Rural patients face nearly double the follow-up wait compared to metro patients.

Rurality	Median (days)	Mean (days)	Sample Size
Metro	15.0	28.9	12,897
Nonmetro	18.0	34.7	94
Rural	**28.0**	**41.2**	**435**

The Journey Duration Paradox

Despite longer follow-up gaps, overall journey *duration* is actually shortest for rural patients. Analysis of the top 10 most common diagnoses across rurality groups reveals why: rural diagnoses skew heavily toward less-specialized conditions handled by family medicine and internal medicine, which require fewer total visits and shorter overall treatment timelines. By contrast, patients are more likely to be diagnosed with complex, specialized conditions at Metro centers (e.g., malignant neoplasms, cardiovascular disease, endocrine disorders) that demand longer multi-visit journeys with frequent follow-ups. The critical insight is that the problem is not journey length - it is that each individual gap between visits is nearly twice as long for rural patients, stretching even routine care timelines unnecessarily.

Ruling Out Alternative Explanations

SVH's Social Determinants of Health survey data reveals that socioeconomic conditions are nearly identical across all rurality groups:

Social Determinant	Metro	Nonmetro	Rural
Transportation barrier	3.9%	3.2%	3.0%
Financial hardship (hard/very hard)	10.7%	10.0%	10.9%
Food insecurity (sometimes/often)	8.3%	7.5%	7.7%
Experienced homelessness	2.2%	1.5%	1.6%

Additionally, **age distribution** is comparable across all three rurality groups, eliminating the hypothesis that disproportionately older rural populations drive longer follow-ups. **MyChart app activation rates** are also similar, though higher adoption correlates with shorter follow-up times. With all demand-side factors ruled out, the evidence points decisively to a **supply-side** cause.

The Specialist Shortage

Provider analysis confirms the supply gap. Metro areas offer 10+ specialties including Orthopedics (900 providers), Hematology/Oncology (754), Endocrinology (725), and Cardiology (493). Rural areas have only 4 specialties: Internal Medicine, Family Medicine, Pediatrics, and OB/GYN. Nonmetro areas are even more constrained with just 3 specialties totaling 93 providers. **Zero cardiologists, oncologists, or endocrinologists practice in rural or nonmetro areas.** Any patient needing specialized care must travel to metro Topeka, which alone houses 800+ providers and the most offices in the SVH network.

Patient-Provider Flow

Analysis of where patients actually receive care reveals the forced migration pattern. Among rural patients, 69.5% of encounters occur with metro providers and only 28.2% with rural providers. Nonmetro patients show an even starker pattern: 29.9% travel to metro providers while 67.7% see rural providers. Metro patients overwhelmingly stay local (96.1% see metro providers). Patients achieve the shortest follow-up gaps when treated by providers matching their own rurality:

Patient → Provider	Metro Provider	Nonmetro Provider	Rural Provider
Metro patient	30 days	36 days	49 days
Nonmetro patient	37 days	55 days	28 days
Rural patient	43 days	43 days	34 days

Only 3% of encounters at rural providers involve same-county patients (vs. 53% for metro), confirming that the healthcare system is centralized in metro Topeka, creating structural delays for the ~32% of the patient population in rural and nonmetro areas.

Key Findings & Recommendations

1. **Lack of specialists in rural and nonmetro regions** forces patients to travel to metro Topeka, creating structural follow-up delays.
2. **Socioeconomic barriers, age distribution, and app usage are similar across groups** — the gap is driven by specialist shortages, not patient-level factors.
3. **Rural journeys are shorter overall but gaps between visits are disproportionately longer** due to less-specialized diagnoses requiring fewer but more widely-spaced encounters.
4. **Expand telehealth and specialist outreach clinics** in rural corridors; prioritize recruitment in cardiology, oncology, and endocrinology; leverage MyChart for cross-county care coordination.

References

Economic Research Service, U.S. Department of Agriculture. (2024). *Rural-urban continuum codes*. https://www.ers.usda.gov/data-products/rural-urban-continuum-codes

Pilafas, J. (2024). *ZIP to FIPS* [Data set]. Kaggle. https://www.kaggle.com/datasets/jonathanpilafas/zip-to-fips